

## 0.1 Butterfly Lemma

**Lemma 0.1.1.** Let  $G$  be a group, and let  $H, K \leq G$  with  $H^* \trianglelefteq H$  and  $K^* \trianglelefteq K$ . Then the following statements hold:

- 1)  $H^*(H \cap K^*) \trianglelefteq H^*(H \cap K)$
- 2)  $K^*(H^* \cap K) \trianglelefteq K^*(H \cap K)$
- 3)  $\frac{H^*(H \cap K)}{H^*(H \cap K^*)} \cong \frac{K^*(H \cap K)}{K^*(H^* \cap K)} \cong \frac{H \cap K}{(H^* \cap K)(H \cap K^*)}$

*Proof.* Since  $H^* \trianglelefteq H$ , for any  $h \in H \cap K$ , we have  $hH^* = H^*h$ . Note that

$$H^*(H \cap K^*) = \bigcup_{h \in H \cap K^*} hH^* = \bigcup_{h \in H \cap K^*} H^*h = (H \cap K^*)H^*$$

Thus,  $H^*(H \cap K^*) \leq H^*(H \cap K)$ . Meanwhile, to show  $H \cap K^* \trianglelefteq H \cap K$ , let  $x \in H \cap K$ . Then

$$x(H \cap K^*)x^{-1} = (xHx^{-1}) \cap (xK^*x^{-1}) = H \cap K^*$$

Similarly,  $H^* \cap K \trianglelefteq H \cap K$ . Since the product of two normal subgroups is normal, we define:

$$L = (H \cap K^*)(H^* \cap K) \trianglelefteq H \cap K.$$

Define a map  $\varphi : H^*(H \cap K) \rightarrow (H \cap K)/L$  by:

$$\varphi(hx) = xL \quad (h \in H^*, x \in H \cap K).$$

**Well-defined:** For  $h_1, h_2 \in H^*$  and  $x_1, x_2 \in H \cap K$ ,

$$h_1x_1 = h_2x_2 \implies h_2^{-1}h_1 = x_2x_1^{-1} \in H^* \cap (H \cap K) = H^* \cap K \leq L.$$

Thus,  $x_2x_1^{-1} \in L \iff x_1L = x_2L$ .

**Homomorphism:** For  $h_1, h_2 \in H^*$  and  $x_1, x_2 \in H \cap K$ ,

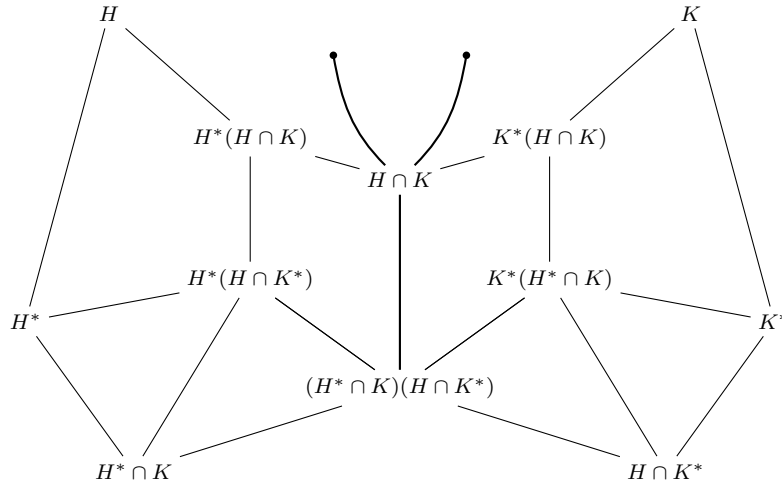
$$\begin{aligned} \varphi((h_1x_1)(h_2x_2)) &= \varphi(h_1(x_1h_2x_1^{-1})x_1x_2) \\ &= \varphi(h_1h_3x_1x_2) \quad (\text{for some } h_3 \in H^* \text{ since } H^* \trianglelefteq H) \\ &= (x_1x_2)L = (x_1L)(x_2L) = \varphi(h_1x_1)\varphi(h_2x_2). \end{aligned}$$

**Onto:** This is trivial as any  $xL \in (H \cap K)/L$  has a pre-image  $1 \cdot x \in H^*(H \cap K)$ .

**Kernel:**

$$\begin{aligned} \ker \varphi &= \{hx \mid h \in H^*, x \in H \cap K, x \in L\} \\ &= \{hx \mid h \in H^*, x \in L\} \\ &= H^*L = H^*(H \cap K^*)(H^* \cap K) \\ &= H^*(H \cap K^*) \end{aligned}$$

Now, by the First Isomorphism Theorem, we obtain the result. □



## 0.2 Subnormal Series of Group

**Definition 0.2.1.** Let  $G$  be a Group.

A Chain of subsets of  $G$   $\{A_i\}_{i=0}^n$  is called *subgroup series* if:

$$1 = A_0 \leq A_1 \leq \cdots \leq A_{n-1} \leq A_n = G$$

**Definition 0.2.2.** Let  $G$  be a Group. A subgroup series  $\{A_i\}_{i=0}^n$  is called *subnormal series* of  $G$  if:

$$1 = A_0 \trianglelefteq A_1 \trianglelefteq \cdots \trianglelefteq A_{n-1} \trianglelefteq A_n = G$$

And for each  $0 \leq i \leq n-1$ ,  $A_{i+1}/A_i$  is called  $i^{th}$  *factor group*.

Specially, if for all  $0 \leq i \leq n-1$ ,  $A_i \trianglelefteq G$ , then it is called *normal series*.

## 0.3 Schreier Refinement Theorem

**Theorem 0.3.1.** Any two subnormal series of a group  $G$  have isomorphic refinements.

*Proof.* Let  $G$  be a group, and let

$$\begin{aligned} \{H_i\}_{i=0}^n : 1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = G \\ \{K_j\}_{j=0}^m : 1 = K_0 \trianglelefteq K_1 \trianglelefteq \cdots \trianglelefteq K_m = G \end{aligned}$$

be two subnormal series for  $G$ . For each  $0 \leq i \leq n-1$  and  $0 \leq j \leq m$ , define:

$$H_{i,j} \stackrel{\text{def}}{=} H_i(H_{i+1} \cap K_j).$$

Then,  $H_{i,j} \trianglelefteq H_{i,j+1}$  for each  $j$  because

$$H_{i,j} = H_i(H_{i+1} \cap K_j) \trianglelefteq H_i(H_{i+1} \cap K_{j+1}) = H_{i,j+1}.$$

(This follows from the fact that  $K_j \trianglelefteq K_{j+1} \implies H_{i+1} \cap K_j \trianglelefteq H_{i+1} \cap K_{j+1}$ ).

In the same sense, for each  $0 \leq j \leq m-1$  and  $0 \leq i \leq n$ , we define:

$$K_{j,i} \stackrel{\text{def}}{=} K_j(K_{j+1} \cap H_i).$$

Now, we can construct refinements for each subnormal series as follows:

$$\begin{array}{ll} H_0 = H_{0,0} \trianglelefteq H_{0,1} \trianglelefteq \cdots \trianglelefteq H_{0,m-1} \trianglelefteq H_{0,m} = H_1 & K_0 = K_{0,0} \trianglelefteq K_{0,1} \trianglelefteq \cdots \trianglelefteq K_{0,n-1} \trianglelefteq K_{0,n} = K_1 \\ H_1 = H_{1,0} \trianglelefteq H_{1,1} \trianglelefteq \cdots \trianglelefteq H_{1,m-1} \trianglelefteq H_{1,m} = H_2 & K_1 = K_{1,0} \trianglelefteq K_{1,1} \trianglelefteq \cdots \trianglelefteq K_{1,n-1} \trianglelefteq K_{1,n} = K_2 \\ \vdots & \vdots \\ H_{n-1} = H_{n-1,0} \trianglelefteq H_{n-1,1} \trianglelefteq \cdots \trianglelefteq H_{n-1,m} = G & K_{m-1} = K_{m-1,0} \trianglelefteq K_{m-1,1} \trianglelefteq \cdots \trianglelefteq K_{m-1,n} = G \end{array}$$

Moreover, the **Butterfly Lemma** gives the following isomorphism: for all  $0 \leq i \leq n-1$  and  $0 \leq j \leq m-1$ ,

$$H_{i,j+1}/H_{i,j} = \frac{H_i(H_{i+1} \cap K_{j+1})}{H_i(H_{i+1} \cap K_j)} \cong \frac{K_j(K_{j+1} \cap H_{i+1})}{K_j(K_{j+1} \cap H_i)} = K_{j,i+1}/K_{j,i}.$$

In summary, the following matrices are Isomorphic elementwise. (Look at the Beautiful Symmetric Relation!!)

$$\begin{bmatrix} H_{0,1}/H_{0,0} & H_{0,2}/H_{0,1} & \cdots & H_{0,m}/H_{0,m-1} \\ H_{1,1}/H_{1,0} & H_{1,2}/H_{1,1} & \cdots & H_{1,m}/H_{1,m-1} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n-1,1}/H_{n-1,0} & H_{n-1,2}/H_{n-1,1} & \cdots & H_{n-1,m}/H_{n-1,m-1} \end{bmatrix} \cong \begin{bmatrix} K_{0,1}/K_{0,0} & K_{0,2}/K_{0,1} & \cdots & K_{0,n}/K_{0,n-1} \\ K_{1,1}/K_{1,0} & K_{1,2}/K_{1,1} & \cdots & K_{1,n}/K_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ K_{m-1,1}/K_{m-1,0} & K_{m-1,2}/K_{m-1,1} & \cdots & K_{m-1,n}/K_{m-1,n-1} \end{bmatrix}^T$$

Consequently, the two subnormal series have isomorphic refinements.  $\square$

## 0.4 Jordan-Hölder Theorem

**Lemma 0.4.1.** Every non-trivial finite group has a maximal normal subgroup.

*Proof.* Let  $G$  be a non-trivial finite group. If  $G$  is a simple group, then its only proper normal subgroup is the trivial group  $\{1\}$ . Thus,  $\{1\}$  is the maximal normal subgroup of  $G$ , and the proof is complete.

Suppose  $G$  is not simple. Then by definition,  $G$  has at least one non-trivial proper normal subgroup  $N$ .

If  $N$  is a maximal normal subgroup of  $G$ , there is nothing to prove.

If  $N$  is not a maximal normal subgroup, then by definition, there exists a proper normal subgroup  $N_1$  of  $G$  such that

$$N < N_1 \triangleleft G$$

If  $N_1$  is not maximal, we can continue this process to obtain a strictly increasing chain of proper normal subgroups:

$$N < N_1 < N_2 < \dots \triangleleft G$$

Since  $G$  is a finite group, the order of these subgroups is bounded by the order of  $G$  (i.e.,  $|N_i| < |G|$ ). Because the size of the subgroups must strictly increase at each step ( $|N| < |N_1| < |N_2| < \dots$ ), this process must terminate in a finite number of steps.

The last subgroup in this terminating chain will be a proper normal subgroup  $M$  such that there is no other normal subgroup between  $M$  and  $G$ . Therefore,  $G$  has a maximal normal subgroup.  $\square$

**Definition 0.4.2.** Let  $G$  be a group. A subnormal series  $\{N_i\}_{i=0}^n$  is called *composition series* if:

For all  $0 \leq i \leq n-1$ ,  $N_{i+1}/N_i$  is a simple group

**Theorem 0.4.3.** Let  $G$  be a Finite Group. Then, the following statements hold:

1.  $G$  has a Composition Series.
2. Every composition series of  $G$  is unique.

More precisely, if  $\{N_i\}_{i=0}^n$  and  $\{M_i\}_{i=0}^m$  are two composition series for  $G$ , then  $n = m$  and there exists a permutation  $\pi$  of  $\{1, 2, \dots, n\}$  such that

$$M_{\pi(i)}/M_{\pi(i)-1} \cong N_i/N_{i-1}$$

In sum,

Every Finite Group has a unique Composition Series

*Proof.* By Lemma, statement 1. is proved:

$G$  has a Maximal normal subgroup  $N_0 \triangleleft G$ , and  $N_0$  has a Maximal normal subgroup  $N_1 \triangleleft N_0$ , this process must terminate in a finite number of steps.

Moreover, every composition series has no refinement when duplications are ignored, and thus the Schreier Refinement Theorem ensures uniqueness.  $\square$

**Definition 0.4.4.** A Group  $G$  is called *solvable* if: There exists a subnormal series  $\{H_i\}_{i=0}^n$  such that

For all  $0 \leq i \leq n-1$ ,  $H_{i+1}/H_i$  is a Abelian Group

**Proposition 0.4.5.** If  $G$  is an Abelian simple group, then  $G \cong \mathbb{Z}_p$  for some prime  $p$ . (The converse is also true.)

*Proof.* Since  $G$  is Abelian, every proper non-trivial subgroup  $1 \subsetneq H \subsetneq G$  is normal. The simplicity of  $G$  implies that  $G$  has no proper non-trivial subgroup. That is, for any non-zero  $g \in G$ , we must have  $\langle g \rangle = G$ .

Now,  $G \cong \mathbb{Z}_k$  for some  $k \in \mathbb{N}$  or  $G \cong \mathbb{Z}$ . If  $G \cong \mathbb{Z}$ , then it has  $2\mathbb{Z}$  as a proper non-trivial subgroup, which contradicts the simplicity of  $G$ . Thus,  $G \cong \mathbb{Z}_n$  for some  $n \in \mathbb{N}$ . If  $n$  is not prime, say  $n = mk$  with  $1 < m < n$ , then it has a proper non-trivial subgroup  $m\mathbb{Z}_n$ , which again contradicts simplicity. Therefore,  $G \cong \mathbb{Z}_p$  for some prime  $p$ .  $\square$

**Lemma 0.4.6.** Let  $G$  be a group and  $N \trianglelefteq G$  be a normal subgroup. Then,

$G/N$  is simple if and only if  $N$  is a maximal normal subgroup of  $G$ .

*Proof.* By the 4th Isomorphism Theorem, we have:

$$\begin{aligned} G/N \text{ is simple} &\iff G/N \text{ has no proper non-trivial normal subgroup} \\ &\iff \text{There is no normal subgroup } H \text{ such that } N \triangleleft H \triangleleft G \\ &\iff N \text{ is a maximal normal subgroup of } G \end{aligned}$$

**Proposition 0.4.7.** Let  $G$  be a finite group. Then,

$G$  is solvable if and only if  $G$  has a composition series such that all its factors are Abelian

*Proof.*  $(\Rightarrow)$  Suppose that  $G$  is solvable. That is,  $G$  has a subnormal series  $\{H_i\}_{i=0}^n$  such that

$$1 = H_0 \triangleleft H_1 \triangleleft \cdots \triangleleft H_{n-1} \triangleleft H_n = G$$

where each factor  $H_{i+1}/H_i$  is Abelian.

For each  $H_{i+1}/H_i$  ( $0 \leq i \leq n-1$ ), since it is a finite Abelian group, it has a composition series  $\{H_{i,j}/H_i\}_{j=0}^{r_i}$  such that

$$1 = H_i/H_i = H_{i,0}/H_i \triangleleft H_{i,1}/H_i \triangleleft \cdots \triangleleft H_{i,r_i}/H_i = H_{i+1}/H_i.$$

by the Jordan-Hölder Theorem. That is, each factors

$$(H_{i,j+1}/H_i)/(H_{i,j}/H_i)$$

are simple. Since  $H_{i+1}/H_i$  is Abelian, these factors are also Abelian.

The 4th Isomorphism Theorem gives a series  $\{H_{i,j}\}_{j=0}^{r_i}$  such that

$$H_i = H_{i,0} \triangleleft H_{i,1} \triangleleft \cdots \triangleleft H_{i,r_i} = H_{i+1}$$

Indeed, all factors of this series are simple and Abelian because the 3rd Isomorphism Theorem gives:

$$(H_{i,j+1}/H_i)/(H_{i,j}/H_i) \cong H_{i,j+1}/H_{i,j}$$

Consequently, by concatenating these series, we obtain

$$1 = H_0 = H_{0,0} \triangleleft H_{0,1} \triangleleft \cdots \triangleleft H_{0,n_0} = H_1 \triangleleft H_{1,1} \triangleleft \cdots \triangleleft H_{n-1,r_{n-1}} = H_n = G$$

which is a composition series satisfying the condition that every factor is Abelian. The converse is trivial.  $\square$

**Corollary 0.4.8.** Let  $G$  be a finite group. Then,

$G$  is solvable if and only if  $G$  has a composition series such that all its factors are cyclic of prime order.